

MTH 166

Lecture-38

Revision of Unit-6 and MCQ Practice

Line Integral w.r.t an Arc Length

Let the parametric representation of curve C be:

$$C: x = x(t), y = y(t), z = z(t); \quad a \leq t \leq b.$$

Let $f(x, y, z)$ be a scalar field function.

Then line integral of scalar function f over curve C w.r.t arc length s is given by:

$$\begin{aligned} I &= \int_C f(x, y, z) ds \\ &= \int_{t=a}^{t=b} \left[f(x(t), y(t), z(t)) \frac{ds}{dt} \right] dt \\ &= \int_{t=a}^{t=b} \left[f(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \right] dt \end{aligned}$$

Line Integral of a Vector Field (Work Done)

Let us consider a vector field $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

We write $\overrightarrow{v(t)} = v_1(t)\hat{i} + v_2(t)\hat{j} + v_3(t)\hat{k}; \quad a \leq t \leq b$

Let curve C is represented by: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

We write $C: \overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$

Then line integral of vector $\vec{v}$ over curve C given by:

$$\begin{aligned} I &= \int_C \vec{v} \cdot d\vec{r} = \int_C \left[\overrightarrow{v(t)} \cdot \frac{d\vec{r}}{dt} \right] dt \\ &= \int_{t=a}^{t=b} \left[(v_1(t)\hat{i} + v_2(t)\hat{j} + v_3(t)\hat{k}) \cdot \left(\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} \right) \right] dt \\ &= \int_{t=a}^{t=b} \left[\left(v_1(t) \frac{dx}{dt} + v_2(t) \frac{dy}{dt} + v_3(t) \frac{dz}{dt} \right) \right] dt \quad \text{This also called as **Work Done**.} \end{aligned}$$

Greens' Theorem:

Let C be a piecewise smooth simple closed curve bounding a region R traced in anticlockwise direction. If f and g are two scalar functions which are continuous and have continuous first order partial derivatives on R , then:

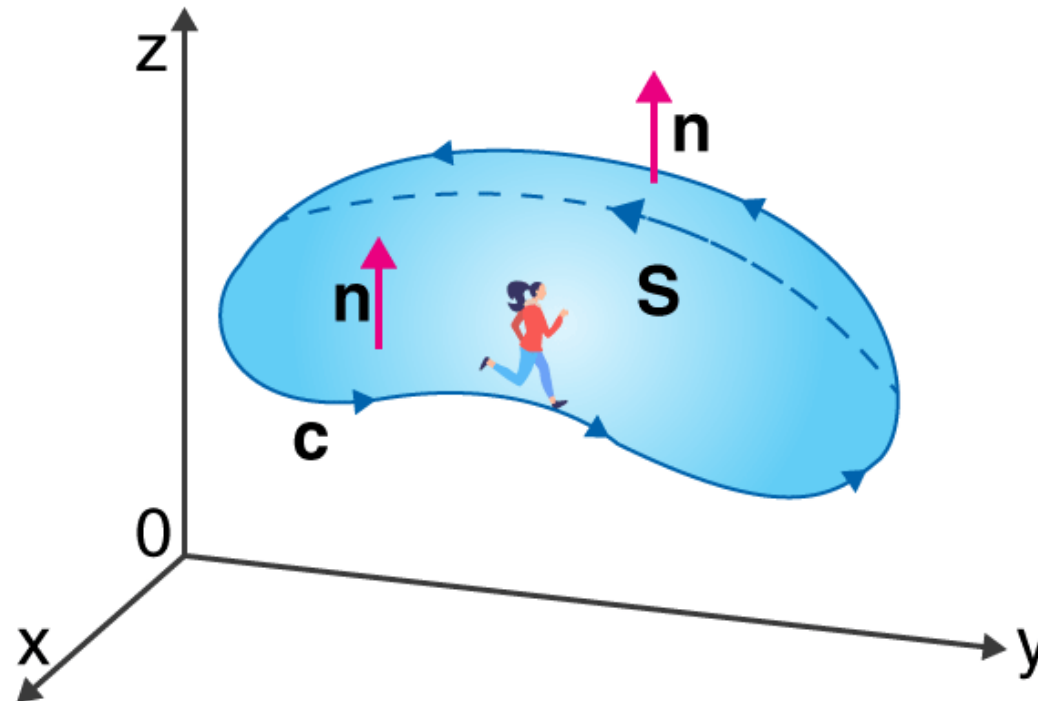
$$\oint_C f(x, y)dx + g(x, y)dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dxdy$$



Stoke's Theorem

Let S be a piecewise smooth orientable surface bounded by a piecewise smooth simple closed curve c traced in a positive direction. Let $\vec{V}$ be a vector which is continuous and have continuous first order partial derivatives. Let $\hat{n}$ is a unit normal vector drawn outward to the surface S . Then:

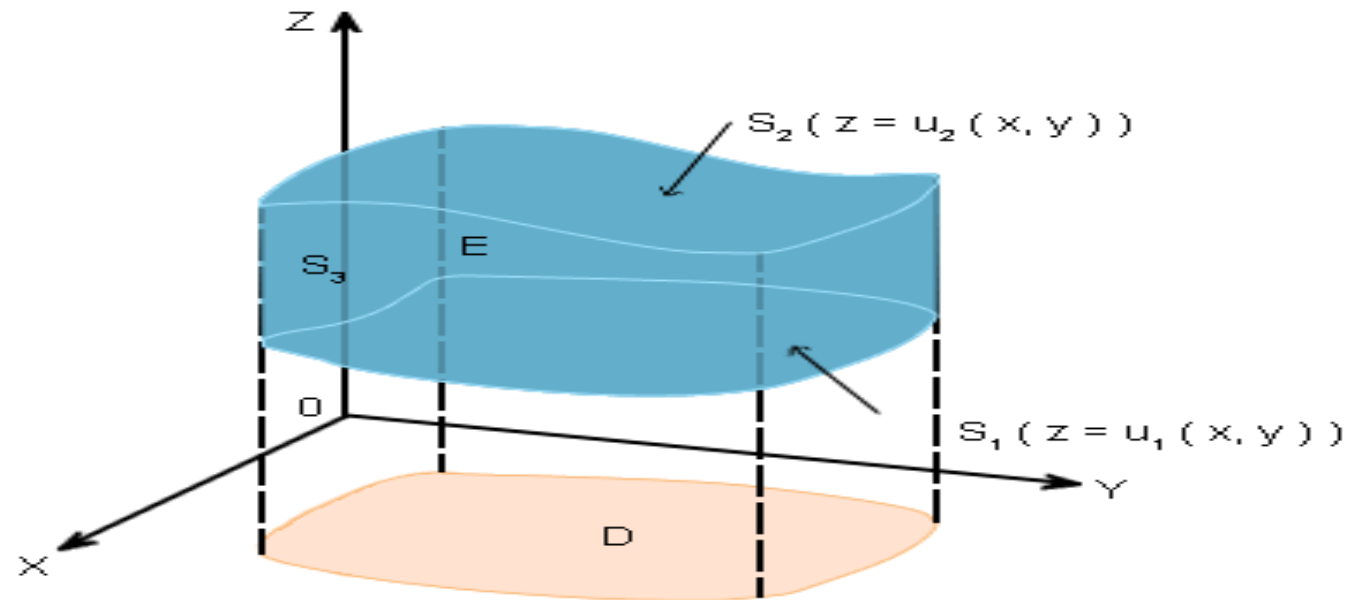
$$\oint_C \vec{V} \cdot d\vec{r} = \int_S \int (\text{Curl } \vec{V}) \cdot \hat{n} dA = \int_S \int (\vec{\nabla} \times \vec{V}) \cdot \hat{n} dA$$



Gauss's Theorem

Let D be a closed and bounded region in 3-dimensional space whose boundary is a piecewise smooth surface S oriented outwards. Let $\vec{V}$ be a vector which is continuous and have continuous first order partial derivatives. Let $\hat{n}$ is a unit normal vector drawn outward to the surface S . Then:

$$\iint_S (\vec{V} \cdot \hat{n}) dA = \iiint_D (\text{div } \vec{V}) dv = \iiint_D (\vec{\nabla} \cdot \vec{V}) dv$$



MCQ Practice Questions

In next section we will be discussing some MCQ questions from previous years question papers.

Q1. If $\vec{v} = (2xyz + 3x^2)\hat{i} + (1 + x^2z)\hat{j} + (x^2y)\hat{k}$ and C is the boundary of the triangle with vertices (1,0,0), (0,2,0), (0,0,3) then integral $\oint_C \vec{V} \cdot \overrightarrow{dr}$ is equal to”

(A) -1

(B) 1

(C) 2

(D) 0

Sol. By Stokes’ theorem: $\oint_C \vec{V} \cdot \overrightarrow{dr} = \int_S \int (\text{Curl } \vec{V}) \cdot \hat{n} dA$

$$\begin{aligned} \text{curl}(\vec{v}) = \vec{\nabla} \times \vec{v} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (2xyz + 3x^2) & (1 + x^2z) & (x^2y) \end{vmatrix} \\ &= \hat{i}(x^2 - x^2) - \hat{j}(2xy - 2xy) + \hat{k}(2xz - 2xz) = 0\hat{i} - 0\hat{j} + 0\hat{k} = \vec{0} \end{aligned}$$

$$\oint_C \vec{V} \cdot \overrightarrow{dr} = \int_S \int (\text{Curl } \vec{V}) \cdot \hat{n} dA = 0$$

Hence, option **(D)** is correct answer.

Q2. If C is any simple smooth and closed curve and $f, g, \frac{\partial f}{\partial y}, \frac{\partial g}{\partial x}$ are continuous in a region R bounded by closed curve C then the value of the integral $\oint_C f dx + g dy$ is:

(A) $\iint_R \left(\frac{\partial f}{\partial y} - \frac{\partial g}{\partial x} \right) dx dy$

(B) $\iint_R \left(\frac{\partial f}{\partial x} - \frac{\partial g}{\partial y} \right) dx dy$

(C) $\iint_R \left(\frac{\partial g}{\partial y} - \frac{\partial f}{\partial x} \right) dx dy$

(D) $\iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$

Sol. By Greens' theorem: $\oint_C f(x, y) dx + g(x, y) dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$

Hence, option **(D)** is correct answer.

Q3. If C is any simple smooth and closed curve and $f, g, \frac{\partial f}{\partial y}, \frac{\partial g}{\partial x}$ are continuous in a region R bounded by closed curve C then which of the following is **Not** representing total area of region R:

$$(A) \oint_C x dy$$

$$\mathbf{(B)} - \oint_C y dx$$

(C) $\oint_C (x dy + y dx)$

$$\textbf{(D)} \iint_R dx dy$$

Sol. By Greens' theorem, area of region R is given by:

$$\text{Area of region R} = \oint_C xdy = -\oint_C ydx = \frac{1}{2} \oint_C (xdy - ydx)$$

Also by double integrals: Area of region R= $\iint_R dx dy$

Hence, option **(C)** is correct answer.

Q4. The value of the integral $\oint_C (y^2 - x^2 + y)dx + (2xy + x)dy$ where C is the closed triangle with vertices (0,0), (2,0), (2,4), taken in order is:

(A) 44

(B) 0

(C) 33

(D) 11

Sol. Comparing it with: $\oint_C f dx + g dy$

$f = (y^2 - x^2 + y)$ implies $\frac{\partial f}{\partial y} = 2y + 1$, $g = (2xy + x)$ implies $\frac{\partial g}{\partial x} = 2y + 1$

By Greens' theorem:

$$\begin{aligned}\oint_C f dx + g dy &= \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy \\ &= \iint_R ((2y + 1) - (2y + 1)) dx dy = 0\end{aligned}$$

Hence, option **(B)** is correct answer.

Q5. If S is the boundary of closed and bounded region D and V is the volume of this region D and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then the value of the surface integral $\iint_S (\vec{r} \cdot \hat{n}) dA$ is:

(A) $3V$

(B) $6V$

(C) $2V$

(D) V

Sol. By Gauss's theorem: $\iint_S (\vec{r} \cdot \hat{n}) dA = \iiint_D (\text{div } \vec{r}) dv$

$$\text{Here, } \text{div}(\vec{r}) = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} = 1 + 1 + 1 = 3$$

$$\text{So, } \iint_S (\vec{r} \cdot \hat{n}) dA = \iiint_D 3 dv = 3 \iiint_D dv = 3(\text{Volume of region } D) = 3V$$

Hence, option **(A)** is correct answer.

Q6. If $\int_P^Q \vec{F} \cdot \overrightarrow{dr}$ is independent of path of integration if:

(A) $\text{curl } (\vec{F}) = 0$

(B) $\text{div } (\vec{F}) = 0$

(C) $\text{grad } (|\vec{F}|) = 0$

(D) None of these

Sol. By Stokes' theorem: $\oint_P^Q \vec{F} \cdot \overrightarrow{dr} = \int_S \int (\text{Curl } \vec{F}) \cdot \hat{n} dA$

So, the integral will be independent of path of integration if $\oint_P^Q \vec{F} \cdot \overrightarrow{dr} = 0$

This is possible if and only if $\text{curl } (\vec{F}) = 0$

Hence, option **(A)** is correct answer.

Q7. The value of the integral $\oint_C ds$ where C is circle of radius 2 units is:

(A) 6π

(B) 8π

(C) 4π

(D) 3π

Sol. The integral $\oint_C ds$ is actually equal to length of arc of curve C.

For a complete closed circle, the length of arc is equal to circumference of circle.

$$\oint_C ds = \text{Circumference of circle} = 2\pi r = 2\pi(2) = 4\pi$$

Hence, option **(C)** is correct answer.

Q8. The value of the integral $\oint_C f(x, y)dy$ where $f(x, y) = x$ and the curve C is $x = 2 \cos t, y = 2 \sin t, 0 \leq t \leq 2\pi$ is :

(A) 6π

(B) 8π

(C) 4π

(D) 3π

Sol. $f(x, y) = x \Rightarrow f(t) = 2 \cos t$

$$y = 2 \sin t \Rightarrow \frac{dy}{dt} = 2 \cos t$$

$$\begin{aligned}\oint_C f(x, y)dy &= \int_{t=0}^{t=2\pi} \left[f(t) \frac{dy}{dt} \right] dt = \int_{t=0}^{t=2\pi} [(2 \cos t) 2 \cos t] dt \\ &= 4 \int_{t=0}^{t=2\pi} \cos^2 t \, dt = 4 \int_{t=0}^{t=2\pi} \left(\frac{1 + \cos 2t}{2} \right) dt = 2 \left[t + \frac{\sin 2t}{2} \right]_{t=0}^{t=2\pi} \\ &= 2 \left[2\pi + \frac{\sin 4\pi}{2} - 0 \right] = 2[2\pi - 0] = 4\pi\end{aligned}$$

Hence, option **(C)** is correct answer.

Q9. The value of the integral $\oint_C (xy^2 - 2y)dx + (x^2y + 3)dy$ where C is the rectangle with vertices $(-1,0), (1,0), (1,1), (-1,1)$ taken in order is:

(A) 24

(B) 4

(C) 10

(D) 0

Sol. Comparing it with: $\oint_C f dx + g dy$

$$f = (xy^2 - 2y) \Rightarrow \frac{\partial f}{\partial y} = 2xy - 2, \quad g = (x^2y + 3) \Rightarrow \frac{\partial g}{\partial x} = 2xy$$

By Greens' theorem:

$$\begin{aligned} \oint_C f dx + g dy &= \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy = \iint_R ((2xy) - (2xy - 2)) dx dy \\ &= 2 \iint_R dx dy = 2 (\text{Area of rectangle with length 2 units and breadth 1 unit}) \\ &= 2(l \times b) = 2(2 \times 1) = 4 \end{aligned}$$

Hence, option **(B)** is correct answer.

Q10. The value of the integral $\oint_C (3xy^2 - 2)dx + (3x^2y + 3)dy$ where C is the simple closed curve, is:

(A) 2

(B) 4

(C) 1

(D) 0

Sol. Comparing it with: $\oint_C f dx + g dy$

$$f = (3xy^2 - 2) \Rightarrow \frac{\partial f}{\partial y} = 6xy, \quad g = (3x^2y + 3) \Rightarrow \frac{\partial g}{\partial x} = 6xy$$

By Greens' theorem:

$$\oint_C f dx + g dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy = \iint_R (6xy - 6xy) dx dy = 0$$

Hence, option **(D)** is correct answer.

Q11. The value of the integral $\oint_C (xy^2 + y)dx + (x^2y)dy$ where C is the closed curve $x^2 + y^2 = 4$

(A) 2π

(B) 4π

(C) -4π

(D) 0

Sol. Comparing it with: $\oint_C f dx + g dy$

$$f = (xy^2 + y) \Rightarrow \frac{\partial f}{\partial y} = 2xy + 1, \quad g = (x^2y) \Rightarrow \frac{\partial g}{\partial x} = 2xy$$

By Greens' theorem:

$$\oint_C f dx + g dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy = \iint_R ((2xy) - (2xy + 1)) dx dy$$

$$= - \iint_R dx dy = (\text{Area of circle } x^2 + y^2 = 4)$$

$$= -(\pi r^2) = -(\pi(2)^2) = -4\pi$$

Hence, option **(C)** is correct answer.



Thanks!